Contents

Hamiltonian Calibration of the Hopf Registry

Toward a Universal Operator Calibration Principle

Companion to the Dirac Operator Calibration Supplement

TuFT / Lambda Framework - July 2026

Abstract

This supplement extends the operator-calibration program developed in the original TuFT work and the subsequent Dirac operator calibration supplement. The prior constructions showed that the Hopf registry may function either as an unscaled relational phase-fiber registry or, once supplied with an operator and a calibrated reference value, as a generator of dimensioned spectral quantities. The Beltrami operator supplied an initial calibration pathway through contact-flow and rotational structure. The Dirac operator supplied a second pathway through spinorial and geometric structure. The present work introduces the Hamiltonian operator as a third pathway, showing that energetic and temporal evolution obey the same underlying calibration pattern.

The central claim does not depend on the Hamiltonian alone. Rather, the Hamiltonian case reveals a more general principle: any registry-compatible operator whose spectrum admits SI interpretation can transport a single calibrated value across the connected spectrum of the Hopf registry by relational ratio. The Hopf registry does not create absolute scale from pure relation. Scale enters through calibration. Once scale enters, the registry propagates that scale across all connected modes determined by the operator.

The resulting principle may be stated schematically as

$$\text{Hopf Registry} + \text{Registry-Compatible Operator} + \text{One SI Calibration} \Longrightarrow \text{Dimensioned Connected Spectrum.}$$

This supplement therefore treats Hamiltonian calibration both as a direct companion to Dirac calibration and as the revealing case from which the broader Universal Operator Calibration Principle becomes explicit.

1. Introduction

The TuFT registry program begins from a distinction between relational structure and dimensioned measurement. The Hopf fibration

$$S^1 \hookrightarrow S^3 \twoheadrightarrow S^2$$

provides a natural model of nested phase-fiber relation. In its unscaled form, the Hopf registry encodes phase, winding, fiber relation, projection, recurrence, and connected modal structure. It does not, by itself, determine an absolute physical value. It can tell how one mode relates to another, but it does not independently specify whether a mode carries one joule, one electronvolt, one hertz, one meter, or any other SI-scaled quantity.

The original TuFT construction used the Beltrami operator to show how a standard operator may convert Hopf-registry structure into a calibrated spectral family. The later Dirac supplement extended this pattern from contact-flow and rotational structure into spinorial geometry. In both cases, the operative structure followed the same form:

$$\text{Unscaled Hopf Registry} + \text{Operator} + \text{One Calibrated Value} \Rightarrow \text{Scaled Spectrum.}$$

The present supplement develops the Hamiltonian case. This case matters because the Hamiltonian occupies a central position in physical theory: it governs time evolution, energy eigenstates, and the dynamical expression of a system. If the Hamiltonian obeys the same registry-calibration pattern as the Beltrami and Dirac operators, then the operator-calibration program no longer appears as a sequence of isolated examples. Instead, it reveals a general structural principle.

The Hamiltonian construction therefore has two purposes. First, it shows directly that Hamiltonian spectra may be calibrated through the Hopf registry in the same formal sense as the Beltrami and Dirac spectra. Given a self-adjoint Hamiltonian H acting on admissible Hopf-registry states,

$$H\psi_n = E_n\psi_n,$$

one calibrated reference energy E_0 fixes every connected energy value by registry ratio:

$$E_n = r_n E_0.$$

Equivalently, using the Planck relation,

$$E_n = \hbar\omega_n,$$

a calibrated reference frequency \omega_0 fixes every connected energetic value whenever

$$\omega_n = r_n\omega_0.$$

Second, the Hamiltonian case makes visible the general operator principle beneath the specific examples. The proof does not depend on a special property unique to the Hamiltonian except its registry-compatible spectral action and its SI-interpretable eigenvalues. The same formal logic applies whenever an operator supplies a connected spectral decomposition over admissible Hopf-registry states and one reference value receives physical calibration.

This leads to the broader statement developed later in this supplement:

A registry-compatible operator with one SI-calibrated reference value determines its connected dimensioned

This statement preserves the discipline of the Lambda framework. It does not claim that pure relation creates scale. It does not claim that the Hopf registry uniquely selects a preferred Hamiltonian. It does not claim that every formal operator automatically carries physical meaning. Rather, it claims that once a compatible operator and one calibrated reference value have been supplied, the Hopf registry transports scale across the connected operator spectrum.

In Lambda terms, the Hopf registry belongs first to the relational or C-frame side of the construction: continuous, phase-based, fibered, and scale-free. SI-calibrated measurement belongs to the L-frame side: discrete, reported, dimensioned, and operationally comparable. The operator mediates between these frames by supplying a spectral witnessing structure. Calibration provides the bridge from relation to measurement. Registry transport then propagates the bridge across connected modes.

2. The Unscaled Hopf Registry and the Necessity of Calibration

The Hopf fibration

$$S^1 \hookrightarrow S^3 \twoheadrightarrow S^2$$

defines a continuous system of nested phase fibers. Within the registry interpretation adopted here, these fibers represent admissible relational states rather than measured quantities. The registry therefore records relationships among modes without independently assigning dimensional magnitude.

Let

$$\mathcal{R} = \{\rho_i\}_{i \in I}$$

denote the Hopf registry. Each registry element \rho_i represents an admissible relational mode, characterized by its position within the connected phase-fiber structure.

Before calibration, the registry determines statements of the form

$$\rho_i \sim \rho_j,$$

or more generally,

$$\rho_i : \rho_j = r_{ij},$$

where r_{ij} is a dimensionless relational ratio. The registry therefore specifies relative structure without specifying absolute measurement.

The Hopf registry may distinguish one mode from another, determine neighborhood relations, preserve topological connectivity, encode linking numbers, transport phase information, and maintain spectral ordering. None of these operations independently assigns an SI quantity to any registry element. Questions such as which state corresponds to one joule, one hertz, one electronvolt, or one meter require the introduction of an external calibration.

This observation should not be interpreted as a deficiency of the registry. Rather, it reflects a familiar distinction appearing throughout mathematics. Coordinate systems determine position but not physical distance until a metric has been supplied. Vector spaces determine direction but not length until an inner product has been chosen. Likewise, probability spaces determine relative likelihoods while remaining independent of the units used to report measured frequencies. In each case, relational structure precedes quantitative calibration.

Accordingly, the TuFT framework separates two logically distinct ingredients: the relational registry \mathcal R, which specifies admissible connectivity and relational organization, and a calibration map \sigma, which assigns one or more registry elements to physically interpreted quantities. Only after these two ingredients have been combined does a dimensioned physical spectrum become available.

Registry Calibration Principle. The Hopf registry determines relational organization but does not independently determine dimensional magnitude. Physical scale enters only through calibration.

$$\boxed{\text{Registry} \neq \text{Measurement.}}$$

$$\boxed{\text{Measurement} = \text{Registry} + \text{Calibration.}}$$

This distinction remains invariant under the choice of compatible operator. Whether one studies rotational flow through the Beltrami operator, spinorial geometry through the Dirac operator, energetic evolution through the Hamiltonian, or another compatible spectral operator, the registry remains unchanged. Only the interpretation of the calibrated spectrum differs.

3. Hamiltonian Construction on the Hopf Registry

Having established the Hopf registry as an unscaled relational substrate, we now introduce the Hamiltonian operator as a dynamical interrogation of that registry. The purpose of this section is constructive rather than interpretive. We seek to show that the Hamiltonian naturally acts upon the same admissible registry states previously considered for the Beltrami and Dirac operators.

Let

$$\mathcal{H}_{\mathrm{R}} = \overline{\mathrm{span}}\{\psi_n\}_{n\in I}$$

denote the Hilbert space generated by admissible Hopf-registry states, where each \psi_n represents a registry-compatible state arising from the relational organization of the Hopf fibration.

The Hamiltonian

$$H : \mathcal{H}_{\mathrm{R}} \to \mathcal{H}_{\mathrm{R}}$$

acts upon these states according to the eigenvalue equation

$$H\psi_n = E_n\psi_n.$$

Because the Hamiltonian is assumed self-adjoint, its spectrum consists of real eigenvalues and an orthogonal decomposition of admissible registry states under the usual hypotheses of the spectral theorem. The present work does not modify these established mathematical properties. Rather, it examines their interpretation within the Hopf registry.

From the registry perspective, each eigenstate corresponds to a particular admissible mode of the common relational substrate. The Hamiltonian does not generate these states from nothing, nor does it replace the registry itself. Instead, it interrogates the registry by assigning an energetic spectrum to already admissible relational modes.

$$\Sigma(H) = \{(E_n, \psi_n)\}_{n\in I}.$$

The Hopf registry determines admissibility. The Hamiltonian determines energetic interpretation. Neither construction subsumes the other.

$$\boxed{\mathcal{R} \overset{\text{admissibility}}{\Longrightarrow} \mathcal{H}_{\mathrm{R}} \overset{H}{\Longrightarrow} \Sigma(H).}$$

At this stage no physical scale has yet entered the construction. The eigenvalues E_n remain purely relational quantities. One may compare them as E_i:E_j or examine their ordering E_1<E_2<\cdots without assigning numerical physical units. Consequently, the Hamiltonian spectrum remains scale-free despite possessing well-defined relational structure.

This observation parallels the previous operator constructions. For the Beltrami operator, rotational and contact-flow modes possessed relational spectral ordering prior to physical calibration. For the Dirac operator, spinorial modes likewise possessed relational ordering before calibration. The Hamiltonian follows exactly the same pattern.

4. Hamiltonian Calibration by Calibration Extension

The previous section established that the Hamiltonian determines a relational spectral decomposition over the admissible states of the Hopf registry. At that stage, the spectrum remained entirely scale-free. The present section introduces calibration and demonstrates that a single calibrated reference value determines the entire connected Hamiltonian spectrum.

4.1 Reference Calibration

Let

$$H\psi_n = E_n\psi_n$$

be the Hamiltonian eigenvalue equation constructed in the previous section. Before calibration, E_n represents only a relational spectral quantity. Suppose now that one admissible registry state \psi_0 receives a physical calibration,

$$E_0 \in \mathbb{R}_{\mathrm{SI}},$$

where \mathbb R_{\mathrm{SI}} denotes the corresponding SI-scaled measurement space. The calibration map therefore consists of

$$\sigma : \psi_0 \mapsto E_0.$$

Only one calibrated value enters the construction.

4.2 Registry Ratios

Because every admissible state belongs to the connected Hopf registry, each eigenvalue possesses a registry relation with the calibrated reference state. Define

$$r_n = \frac{E_n}{E_0},$$

where r_n is dimensionless. The registry therefore determines the complete family

$$\{r_n\}_{n \in I}.$$

The quantity r_n contains relational information only. It introduces no additional physical scale.

4.3 Calibration Extension

The calibrated value E_0 extends naturally over the connected spectrum by registry relation,

$$E_n = r_n E_0.$$

This operation shall be called Calibration Extension. Calibration Extension assigns physical dimension to every connected registry mode while preserving the relational structure already determined by the operator. No additional calibration is required. No second physical constant enters. The remaining spectrum inherits scale through registry connectivity.

4.4 SI Interpretation

Suppose the calibrated observable represents energy. Then

$$[E] = \mathrm{J}.$$

Using the standard relation

$$E = \hbar\omega,$$

the calibrated reference value satisfies

$$E_0 = \hbar\omega_0.$$

If the registry determines

$$\omega_n = r_n \omega_0,$$

then

$$E_n = \hbar\omega_n = \hbar r_n \omega_0 = r_n E_0.$$

Thus frequency calibration and energy calibration become equivalent descriptions of the same registry extension. The Planck relation introduces no new relational information. It merely converts one SI interpretation into another.

4.5 Hamiltonian Calibration Theorem

Theorem 1: Hamiltonian Calibration. Let \mathcal R be a connected Hopf registry, \mathcal H_{\mathrm R} its associated Hilbert space of admissible states, and H a registry-compatible self-adjoint Hamiltonian. Suppose H\psi_n=E_n\psi_n admits a connected spectral decomposition. If one reference eigenvalue E_0 receives SI calibration, then every connected eigenvalue receives a unique SI value through Calibration Extension, E_n=r_nE_0, where r_n is the corresponding dimensionless registry ratio.

$$\boxed{(\mathcal{R}, H, \sigma) \Longrightarrow \Sigma_{\mathrm{SI}}(H).}$$

Proof. The Hopf registry determines connectivity among admissible states. The Hamiltonian determines the relational spectrum over those states. Calibration fixes one reference eigenvalue. Because the remaining eigenvalues already possess registry ratios relative to the calibrated reference, multiplication by the corresponding dimensionless ratio assigns an SI value to every connected mode. No further calibration enters the construction. Therefore the calibrated spectrum follows uniquely from the registry ratios and the chosen reference value.

□

4.6 Structural Interpretation

The significance of the theorem lies less in the Hamiltonian itself than in the structure of the proof. The proof nowhere relies upon a property unique to energetic evolution. Instead, it depends only upon four ingredients: a connected relational registry, an admissible operator acting on that registry, a relational spectral decomposition, and one SI-calibrated reference value.

5. Registry-Compatible Operators

The Hamiltonian construction establishes a calibration mechanism formally identical to those previously obtained for the Beltrami and Dirac operators. This observation suggests that the common mathematical structure resides neither in the individual operators nor in their physical interpretations, but in the manner by which they interrogate the underlying Hopf registry.

5.1 The Registry as the Primary Object

Within the TuFT framework, the Hopf registry occupies the primary structural position. It determines the admissible relational organization of states independently of any particular physical observable. Consequently, the registry exists prior to any specific operator acting upon it.

Operators do not generate the registry. They do not modify its relational organization. They do not replace its topological structure. Instead, operators act upon admissible registry states to reveal particular measurable aspects of the same underlying relational object.

5.2 Definition: Registry-Compatible Operator

Definition 1: Registry-Compatible Operator. A linear operator A:\mathcal H_{\mathrm R}\rightarrow\mathcal H_{\mathrm R} shall be called registry-compatible whenever its domain consists of admissible Hopf-registry states; its action admits a spectral decomposition over those states; the resulting spectrum preserves the connected relational organization induced by the Hopf registry; and its eigenvalues admit interpretation as measurable quantities after calibration.

No assumption concerning the specific observable enters this definition. The observable may represent energy, frequency, momentum, angular momentum, curvature, spin, flow, or another measurable quantity. The definition concerns only the structural relation between the operator and the registry.

5.3 Operators as Registry Witnesses

When a registry-compatible operator acts on the Hopf registry, it may be said to witness one measurable aspect of the common relational substrate. This use of witness should be understood descriptively rather than as a separate technical object.

Beltrami witnesses rotational and contact-flow organization. Dirac witnesses spinorial and geometric organization. Hamiltonian witnesses energetic and temporal organization. The underlying registry remains unchanged throughout all three constructions. Only the witnessed observable differs.

$$\mathcal{R} \overset{B}{\Longrightarrow} \Sigma(B), \quad \mathcal{R} \overset{D}{\Longrightarrow} \Sigma(D), \quad \mathcal{R} \overset{H}{\Longrightarrow} \Sigma(H).$$

Each operator produces its own spectral decomposition. Each decomposition admits independent SI calibration. Each calibration extends over the connected spectrum according to the same relational mechanism established previously. Consequently, the calibration procedure appears independent of the specific operator. It depends instead upon the common registry structure.

6. The Operator Calibration Principle

The preceding sections developed three constructions based upon the same underlying Hopf registry. The Beltrami operator, the Dirac operator, and the Hamiltonian each produced a spectral decomposition over admissible registry states. In every case, the resulting spectrum remained relational until one reference value received physical calibration. Once calibration entered, the remaining spectrum acquired dimensioned interpretation through the connected relational structure of the registry.

6.1 Structural Ingredients

The developments reveal four ingredients necessary for registry calibration: a connected relational registry \mathcal R; a registry-compatible operator A:\mathcal H_{\mathrm R}\rightarrow\mathcal H_{\mathrm R}; a spectral decomposition A\psi_n=\lambda_n\psi_n whose eigenvalues preserve the relational organization inherited from the registry; and one reference eigenvalue \lambda_0 receiving physical calibration through an externally specified SI value.

6.2 Operator Calibration Principle

Theorem 2: Operator Calibration Principle. Let \mathcal R be a connected Hopf registry. Let A be a registry-compatible self-adjoint operator acting upon the Hilbert space of admissible registry states. Suppose A\psi_n=\lambda_n\psi_n admits a connected spectral decomposition. Suppose further that one reference eigenvalue \lambda_0 receives SI calibration. Then every registry-connected eigenvalue receives a uniquely determined SI value through calibration extension, \lambda_n=r_n\lambda_0, where r_n denotes the corresponding dimensionless registry ratio.

$$\boxed{(\mathcal{R}, A, \sigma) \Longrightarrow \Sigma_{\mathrm{SI}}(A).}$$

Proof. The registry determines the admissible relational organization of states. The operator determines a spectral decomposition over those states. Because the spectrum remains connected through registry relations, each eigenvalue possesses a unique dimensionless ratio relative to the calibrated reference eigenvalue. Multiplication by that calibrated value therefore assigns an SI interpretation to every connected eigenvalue while preserving the operator-induced spectral organization. No additional calibration enters the construction. Accordingly, calibration extends uniquely over the connected spectrum.

□

6.3 Immediate Corollaries

Corollary 1: Beltrami. The Beltrami operator satisfies the hypotheses of the Operator Calibration Principle. Consequently, one calibrated Beltrami eigenvalue determines the connected Beltrami spectrum by calibration extension.

Corollary 2: Dirac. The Dirac operator satisfies the hypotheses of the Operator Calibration Principle. Consequently, one calibrated Dirac eigenvalue determines the connected Dirac spectrum by calibration extension.

Corollary 3: Hamiltonian. The Hamiltonian satisfies the hypotheses of the Operator Calibration Principle. Consequently, one calibrated Hamiltonian eigenvalue determines the connected Hamiltonian spectrum by calibration extension.

6.4 Lambda Interpretation

Within the Lambda framework, the Operator Calibration Principle may be understood as a bridge between relational and measured description. The Hopf registry supplies a continuous relational organization independent of physical units. The registry-compatible operator interrogates that organization through a chosen observable. Calibration introduces an externally specified measurement scale. Calibration extension then carries that scale across the connected spectrum without altering the underlying relational organization.

The resulting construction distinguishes three logically independent roles: the registry organizes; the operator interrogates; the calibration measures. None of these functions replaces the others. Together they produce a dimensioned spectral description of the same underlying relational substrate.

7. Beltrami, Dirac, and Hamiltonian as Operator Realizations

The Operator Calibration Principle reframes the previous constructions. The Beltrami, Dirac, and Hamiltonian cases should no longer be read as independent demonstrations requiring separate justification at the level of first principle. Instead, each should be read as a realization of one shared calibration architecture.

$$\text{Registry} + \text{Compatible Operator} + \text{One Calibrated Value} \Longrightarrow \text{Connected Dimensioned Spectrum.}$$

7.1 Beltrami Realization

The Beltrami operator enters the original TuFT construction through rotational, contact-flow, and helicity-related structure. Its eigenvalue equation may be written schematically as

$$B\psi_n = \beta_n\psi_n.$$

In the unscaled registry, the quantities \beta_n possess relational meaning only. They order and distinguish admissible flow modes, but they do not independently determine physical magnitude. Once one reference value \beta_0 receives calibration, every connected Beltrami eigenvalue receives scale through

$$\beta_n = r_n\beta_0.$$

Thus the Beltrami construction expresses the calibration of rotational or contact-flow structure.

7.2 Dirac Realization

The Dirac operator extends the same calibration architecture into spinorial geometry. Its eigenvalue equation may be written

$$D\psi_n = \mu_n\psi_n.$$

The values \mu_n encode a spinorial or geometric spectral structure over the admissible registry states. Prior to calibration, they remain relational. Given one calibrated reference value \mu_0, the connected spectrum receives dimensioned interpretation through

$$\mu_n = r_n\mu_0.$$

The Dirac construction therefore shows that the same registry-calibration mechanism applies beyond contact-flow geometry. It applies also to spinorial and geometric modes.

7.3 Hamiltonian Realization

The Hamiltonian realization developed in this supplement extends the same pattern into energetic and temporal dynamics. The Hamiltonian eigenvalue equation is

$$H\psi_n = E_n\psi_n.$$

Before calibration, the spectrum \{E_n\} possesses relational order and ratio but no absolute energetic scale. Once one reference value E_0 receives SI interpretation, every connected energetic value follows by

$$E_n = r_nE_0.$$

Equivalently, if E_0=\hbar\omega_0 and \omega_n=r_n\omega_0, then

$$E_n = \hbar\omega_n = r_n E_0.$$

The Hamiltonian therefore confirms that the same calibration architecture applies to dynamical evolution and energy spectra.

7.4 Comparative Summary

Beltrami operator. Spectral form: B\psi_n=\beta_n\psi_n. Calibrated reference value: \beta_0. Registry role: flow-mode decomposition. Physical interpretation: contact flow, helicity, and rotation.

Dirac operator. Spectral form: D\psi_n=\mu_n\psi_n. Calibrated reference value: \mu_0. Registry role: spinorial decomposition. Physical interpretation: spin geometry, mass-like scale, and chirality.

Hamiltonian operator. Spectral form: H\psi_n=E_n\psi_n. Calibrated reference value: E_0. Registry role: energy-mode decomposition. Physical interpretation: energy, time evolution, and dynamics.

The comparison should not be read as claiming that these operators possess identical physical meaning. They do not. Rather, it shows that they share a common calibration architecture when interpreted over a connected Hopf registry.

7.5 Consequence for Future Operators

Future operators may enter the framework whenever they satisfy the same registry-compatible conditions. Possible candidates include Laplace-type operators, Hodge-Laplacian operators, momentum operators, angular momentum operators, Casimir operators, Maxwell-type operators, and gauge-covariant operators.

Each such extension must specify the registry domain, specify the operator, establish registry compatibility, identify the spectral decomposition, introduce exactly one calibrated reference value for each connected component, derive the connected dimensioned spectrum by calibration extension, and avoid claiming that the registry alone creates scale. This gives the operator program a repeatable method rather than a collection of analogies.

8. Stopping-State Discipline and Non-Claims

The preceding sections establish a Hamiltonian calibration construction and abstract it into an Operator Calibration Principle. This section states the stopping conditions explicitly. These limits matter because the construction becomes stronger, not weaker, when its scope remains precise.

8.1 Established Claim

The present work establishes the following claim:

$\boxed{\text{Given a connected Hopf registry, a registry-compatible operator, and one SI-calibrated reference value, the c}}$

Equivalently,

$$(\mathcal{R}, A, \sigma) \Longrightarrow \Sigma_{\mathrm{SI}}(A),$$

provided that \mathcal R supplies the connected relational registry, A acts on admissible registry states, A admits a connected spectral decomposition, one reference value \lambda_0 receives SI calibration, and the ratios r_n remain well-defined across the connected spectrum.

8.2 Non-Claim: Scale Does Not Emerge from Pure Relation

The construction does not claim that scale emerges from pure relation. The unscaled Hopf registry supplies ratios, connectivity, and admissible relational structure. It does not independently supply joules, hertz, meters, electronvolts, or any other SI unit. Scale must enter through calibration.

$$\boxed{\text{Relation transports scale after calibration; relation does not create scale before calibration.}}$$

8.3 Non-Claim: The Registry Does Not Select a Unique Operator

The construction does not claim that the Hopf registry uniquely selects the Hamiltonian, the Dirac operator, the Beltrami operator, or any other operator. Different registry-compatible operators may act on the same underlying registry and expose different measurable aspects of it.

$$\boxed{\text{The registry supplies admissible structure; the operator supplies a chosen spectral interrogation.}}$$

8.4 Non-Claim: Not Every Formal Operator Qualifies

The construction does not claim that every formal operator automatically participates in the calibration principle. An operator must satisfy registry compatibility. It must act on admissible registry states, admit an appropriate spectral decomposition, preserve connected registry relations, and permit physical interpretation after calibration. Thus the general claim applies only to operators satisfying the stated hypotheses.

8.5 Non-Claim: Connectedness Cannot Be Omitted

The connectedness condition remains essential. If an operator spectrum decomposes into disconnected components, a calibration assigned to one component does not automatically determine the scale of another disconnected component. In such a case, each disconnected component may require its own calibration anchor.

$$\boxed{\text{One calibrated value fixes one connected spectral component.}}$$

8.6 Lambda Stopping State

In Lambda terms, this work stops at a bridge-supported claim. It establishes that the Hamiltonian participates in the same calibration pattern as Beltrami and Dirac, provided the operator satisfies registry compatibility and one SI-scaled value enters as calibration. It does not claim absolute frame-independent closure. It does not claim that the C-frame registry internally produces L-frame measurement without residue.

$$\boxed{\text{C-frame relational registry} + \text{operator witness} + \text{L-frame calibration} \Rightarrow \text{dimensioned connected spectrum.}}$$

This constitutes the proper stopping state of the present supplement.

9. Conclusion

The present supplement developed the Hamiltonian calibration of the Hopf registry as a direct companion to the Dirac operator calibration work and the original Beltrami-based TuFT construction. The Hamiltonian case demonstrates that energetic and temporal spectra follow the same operator-calibration pattern previously observed in flow and spinorial settings.

The central result may be stated succinctly: the Hopf registry supplies relational organization; a registry-compatible operator supplies spectral decomposition; one calibrated SI reference value supplies physical scale; and Calibration Extension propagates that scale across the connected spectrum.

$$\boxed{\mathcal{R} + A + \sigma \Longrightarrow \Sigma_{\mathrm{SI}}(A).}$$

This result does not assert that the registry creates scale from pure relation. It does not assert that the Hopf registry uniquely selects a physical operator. It does not assert that every formal operator carries physical meaning. It states a disciplined conditional principle: whenever a connected Hopf registry, a registry-compatible operator, and one calibrated reference value are specified, the connected dimensioned spectrum follows by relational calibration.

Beltrami, Dirac, and Hamiltonian therefore appear as the first three realizations of a common operator-calibration architecture. The present work proposes this architecture as a reusable method for future TuFT/Lambda investigations of Laplacian, Hodge, momentum, angular momentum, Casimir, Maxwell-type, and gauge-covariant operators. Each future extension must satisfy the same stopping discipline: define the registry, specify the operator, establish compatibility, identify connected spectral ratios, introduce calibration, and derive only the corresponding connected dimensioned spectrum.

The final claim of this supplement is therefore precise:

Hamiltonian calibration works in the same formal sense as Beltrami and Dirac calibration; more generally, e

All stronger claims require additional premises and remain outside the scope of the present work.